

Linear Algebra 2nd Test

1. Find the determinant of the given matrix. (10%)

$$\begin{vmatrix} 1 & 0 & 4 & 0 & 0 \\ -1 & 0 & 1 & 1 & 0 \\ 2 & 0 & 3 & 0 & 0 \\ 0 & 1 & 0 & 4 & 3 \\ -2 & 0 & 0 & 1 & -1 \end{vmatrix}$$

2. (16%)

Let $T : P_3 \rightarrow P_3$ be the linear transformation defined by $T(p(x)) = D^2(p(x)) - 4D(p(x)) + p(x)$, where $B = B' = (2x, 1 - x, x + x^2, x^3)$, and D represents differentiation.

- (1) Find the matrix representation A of linear transformation T relative to the bases B and B' .
 (2) Use the matrix A to compute $T(-x^3)$.

3. Find the coordinate vector of the given vector relative the indicated ordered basis
 $4x^3 + 3x^2 + 2x + 1$ in P_3 relative to $(1, 1+x, 1+x+x^2, 1+x+x^2+x^3)$ (10%)

4.

(12%) Two linear transformations T, T' are defined as follow:

$$T : R^2 \rightarrow R^3 \text{ 為 } T([x_1, x_2]) = [2x_1 + x_2, x_1, x_1 - x_2]$$

$$T' : R^3 \rightarrow R^2 \text{ 為 } T'([x_1, x_2, x_3]) = [x_1 - x_2 + x_3, x_1 + x_2]$$

- (1) Find the standard matrix representation for the $T \circ T'$ that carries R^2 into R^2 .
 (2) Find a formula for $(T \circ T')([x_1, x_2])$

5. Find the volume of the box having the given vectors as adjacent edges. (8%)
 $3\mathbf{i} + 6\mathbf{j} - 4\mathbf{k}, -4\mathbf{i} - 2\mathbf{k}, -\mathbf{i} + \mathbf{j} + 4\mathbf{k}$

6.(10%)

Let $W_1 = \text{sp}((6, 4, 3), (1, 1, 1))$ and $W_2 = \text{sp}((3, 1, 0), (1, 1, 0))$ in $\mathbb{R}^3$. Find a set of generating vectors for $W_1 \cap W_2$.

7. (16%)是非題：非者須說明理由(True or False: Give reasons)

- (a) A linear transformation $T: V \rightarrow V'$ carries the zero vector of V into the zero vector of V' .
- (b) Distinct vectors in finite-dimensional vector space V have distinct coordinate vectors relative to the same ordered basis B for V .
- (c) Two subspaces of a vector space V may have empty intersection.
- (d) There is exactly one isomorphism $T: P_{10} \rightarrow R^{11}$.
- (e) In order for the determinant of a 3×3 matrix to be zero, two rows of the matrix must be parallel vectors in R^3 .
- (f) If the matrix A is multiplied by a scalar c , the determinant of the resulting matrix is $c \cdot \det(A)$.
- (g) The defined function $T([x_1, x_2]) = [2x_1 + x_2, x_1 - 1, x_1 - x_2]$ is a linear transformation from R^2 into R^3 .
- (h) The following subsets of $M_{2 \times 3}$ form subspaces

The subsets of matrices of the form $\begin{bmatrix} a & 2a & 3a \\ b & 2b & 3b \end{bmatrix}$

8. (8%) Find the adjoint of the matrix A (i.e. $\text{adj}(A)$)

$$A = \begin{bmatrix} 6 & 2 & 8 \\ -3 & 4 & 1 \\ 4 & -4 & 5 \end{bmatrix}$$

9. (10%)

Decide whether or not the set R^2 , together with the following operations

$$[x, y] + [t, s] = [x+t, 2y+2s], r[x, y] = [rx, ry],$$

is a vector space.